

Setup Nodemailer untuk Email OTP

Status Saat Ini

- ☑ Nodemailer sudah di-install (`npm install nodemailer`)
- ☑ Email service sudah dibuat (`src/server/services/email.service.ts`)
- ⚠ Environment variables belum configured

Langkah Setup

1. Pilih SMTP Provider

Option A: Gmail (Recommended untuk testing)

Kelebihan: Gratis, mudah setup

Kekurangan: Limited untuk production (low rate limit)

Setup steps:

1. Buka <https://myaccount.google.com/security>
2. Enable **2-Step Verification**
3. Buka <https://myaccount.google.com/apppasswords>
4. Pilih **Mail** dan **Windows Computer** (atau device lain)
5. Generate password baru
6. Copy app password yang digenerate

.env.local configuration:

```
NODE_ENV=development

SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=your-email@gmail.com
SMTP_PASSWORD=xxxx-xxxx-xxxx-xxxx # App password yang digenerate
SMTP_FROM_EMAIL=noreply@desasangkima.com
```

Option B: SendGrid (Production-grade)

Kelebihan: Professional, reliable, scalable

Kekurangan: Perlu signup, paid (tapi ada free tier)

Setup steps:

1. Daftar di <https://sendgrid.com>

2. Verify sender email
3. Create API key di Settings → API Keys
4. Copy API key

.env.local configuration:

```
NODE_ENV=production

SMTP_HOST=smtp.sendgrid.net
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=apikey
SMTP_PASSWORD=SG.xxxxxx # Paste API key
SMTP_FROM_EMAIL=noreply@desasangkima.com
```

Option C: Mailtrap (Development/Testing)

Kelebihan: Perfect untuk testing, free tier bagus

Kekurangan: Tidak untuk production (fake SMTP)

Setup steps:

1. Daftar di <https://mailtrap.io>
2. Setup inbox
3. Copy SMTP credentials dari "Show Credentials"

.env.local configuration:

```
NODE_ENV=development

SMTP_HOST=live.smtp.mailtrap.io
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=your-username@mailtrap.io
SMTP_PASSWORD=your-password
SMTP_FROM_EMAIL=noreply@desasangkima.com
```

2. Update .env.local

Pilih salah satu provider di atas dan update **.env.local** Anda:

```
# Database
DB_HOST=localhost
DB_PORT=3306
```

```
DB_USER=root
DB_PASSWORD=
DB_NAME=db_websangkima

# JWT
JWT_SECRET=your-secret-key

# Node Environment
NODE_ENV=development

# Email SMTP (Pilih salah satu provider)
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=your-email@gmail.com
SMTP_PASSWORD=your-app-password
SMTP_FROM_EMAIL=noreply@desasangkima.com
```

3. Test Email Configuration

Create test file:

```
// test-email.ts
import { sendOTPEmail } from './src/server/services/email.service';

async function test() {
  try {
    await sendOTPEmail("your-test-email@gmail.com", "1234");
    console.log("✅ Email sent successfully!");
  } catch (error) {
    console.error("❌ Email failed:", error);
  }
}

test();
```

Run:

```
npx tsx test-email.ts
```

Development Mode vs Production Mode

🔑 Development (NODE_ENV=development)

```
NODE_ENV=development
```

Behavior: OTP ditampilkan di **console**, tidak dikirim email

Gunakan untuk: Local testing, development environment

Console output:

```
✉ OTP Email to user@example.com:
  OTP Code: 1234
  Valid for 15 minutes
```

🚀 Production (NODE_ENV=production)

```
NODE_ENV=production
SMTP_HOST=...
SMTP_USER=...
SMTP_PASSWORD=...
```

Behavior: Email benar-benar dikirim via SMTP

Gunakan untuk: Staging, production environment

How Email Service Works

File: `src/server/services/email.service.ts`

```
// Development Mode
if (process.env.NODE_ENV !== "production") {
  console.log(`✉ OTP Email to ${to}:`);
  console.log(`  OTP Code: ${otp}`);
  return;
}

// Production Mode - Send via SMTP
const transporter = nodemailer.createTransport({
  host: process.env.SMTP_HOST,
  port: parseInt(process.env.SMTP_PORT || "587"),
  secure: process.env.SMTP_SECURE === "true",
  auth: {
    user: process.env.SMTP_USER,
    pass: process.env.SMTP_PASSWORD,
  },
});

await transporter.sendMail({...});
```

Testing Email Flow

Step 1: Register User

```
curl -X POST http://localhost:3000/api/auth/register \
-H "Content-Type: application/json" \
-d '{
  "name": "Test User",
  "email": "test@example.com",
  "nik": "1234567890123456",
  "password": "password123"
}'
```

Development Response (console log):

```
📧 OTP Email to test@example.com:
  OTP Code: 1234
  Valid for 15 minutes
```

Production Response (email sent):

- Email terkirim ke inbox
- User bisa check email untuk OTP

Step 2: Verify OTP

```
curl -X POST http://localhost:3000/api/auth/verify-otp \
-H "Content-Type: application/json" \
-d '{
  "userId": "user_id_dari_register",
  "otp": "1234"
}'
```

Step 3: Login

```
curl -X POST http://localhost:3000/api/auth/login \
-H "Content-Type: application/json" \
-d '{
  "email": "test@example.com",
  "password": "password123"
}'
```

Troubleshooting

✗ Email tidak terkirim di production

Solusi:

1. Cek `.env` - SMTP_HOST, USER, PASSWORD benar?
2. Cek `NODE_ENV=production`
3. Check firewall - port 587 terbuka?
4. Try different SMTP provider
5. Check email provider logs (Gmail: myaccount.google.com/security-check)

✗ ECONNREFUSED pada console

Penyebab: Nodemailer mencoba connect SMTP tapi server down

Solusi:

- Cek SMTP_HOST dan PORT benar
- Cek internet connection
- Try Mailtrap untuk testing

✗ Authentication failed

Penyebab: Username/password salah

Solusi:

- Re-generate app password (untuk Gmail)
- Double-check credentials di provider
- Pastikan tidak ada extra spaces

i Development mode - hanya console log

Expected behavior - tidak ada bug

Set `NODE_ENV=production` untuk real email

Recommended Setup

Untuk Development

```
NODE_ENV=development
```

- ☒ Tidak perlu SMTP credentials
- ☒ Lihat OTP langsung di console
- ☒ Perfect untuk local testing

Untuk Production

```
NODE_ENV=production
SMTP_HOST=smtp.sendgrid.net
SMTP_PORT=587
SMTP_SECURE=false
SMTP_USER=apikey
SMTP_PASSWORD=SG.xxxxx
SMTP_FROM_EMAIL=noreply@desasangkima.com
```

- ☒ Real email service
 - ☒ Professional, scalable
 - ☒ Good reliability
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Apa Pilihan Anda?

1. **Gmail** - Simple, free, bagus untuk dev
2. **SendGrid** - Professional, production-ready
3. **Mailtrap** - Testing only, fake SMTP

Rekomendasikan mana yang mau digunakan?